

Loxahatchee River-Lake Worth Creek Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

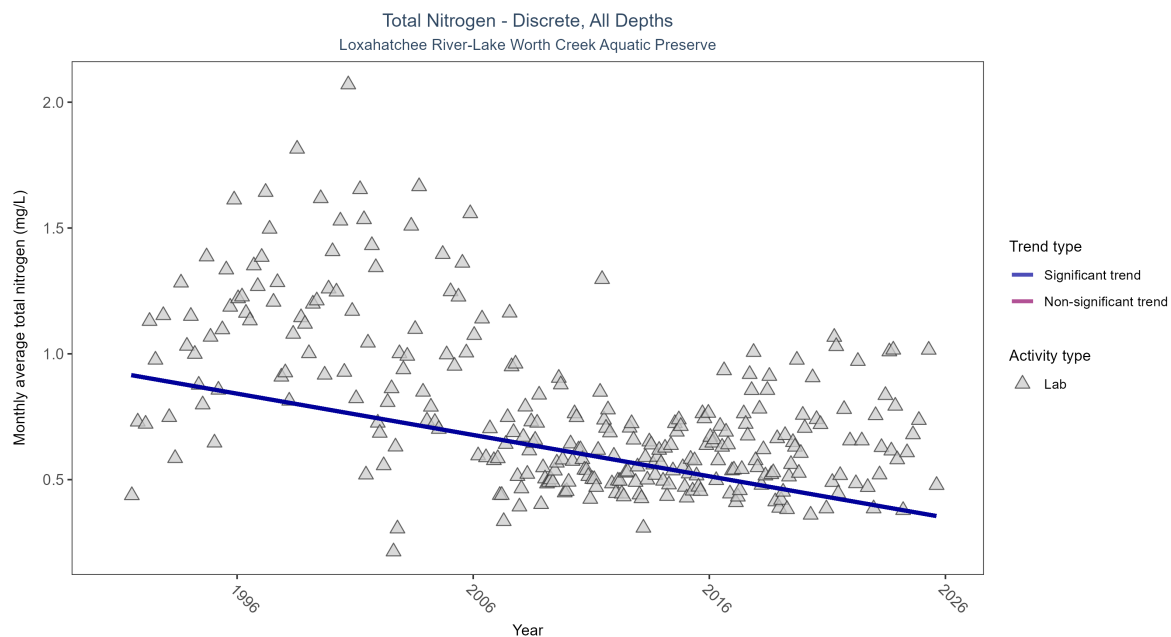


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen decreased by 0.02 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	4775	35	1991 - 2025	0.706	-0.27174	0.92384	-0.01642	0

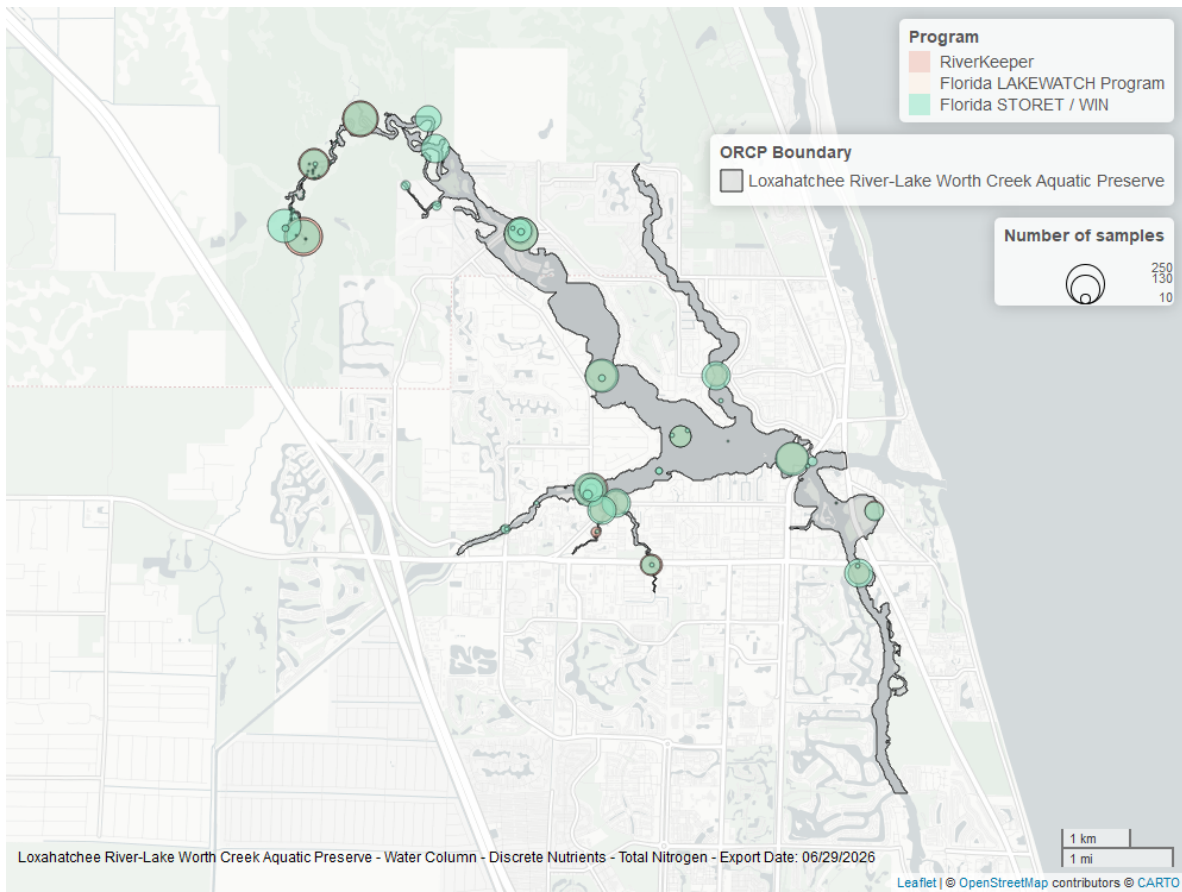


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

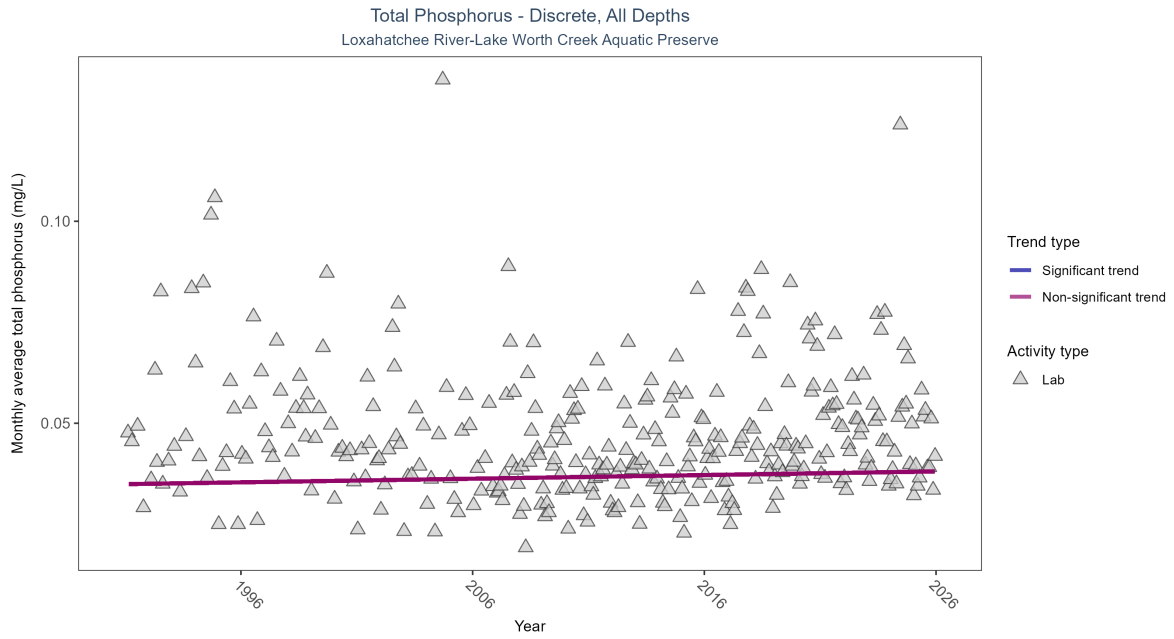


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1991 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6759	35	1991 - 2025	0.042	0.06964	0.03492	0.00009	0.2154

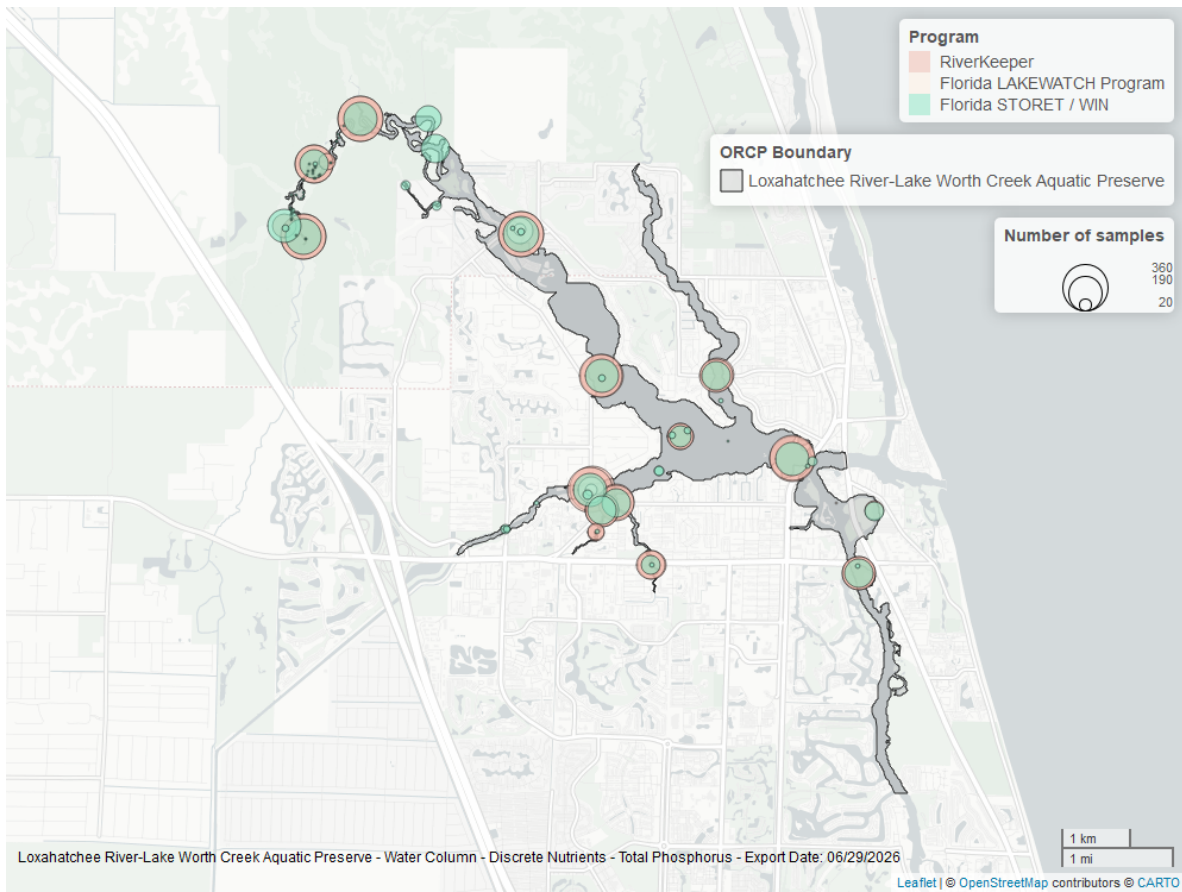


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Discrete

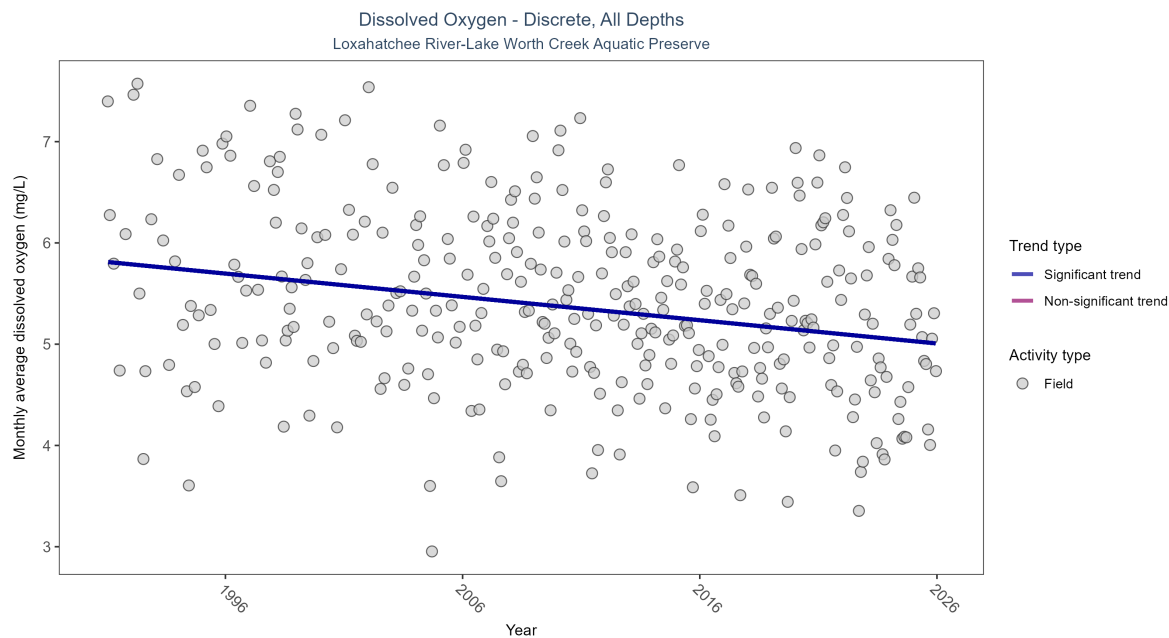


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Monthly average dissolved oxygen decreased by 0.02 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	13210	35	1991 - 2025	5.59	-0.26282	5.81215	-0.02301	0

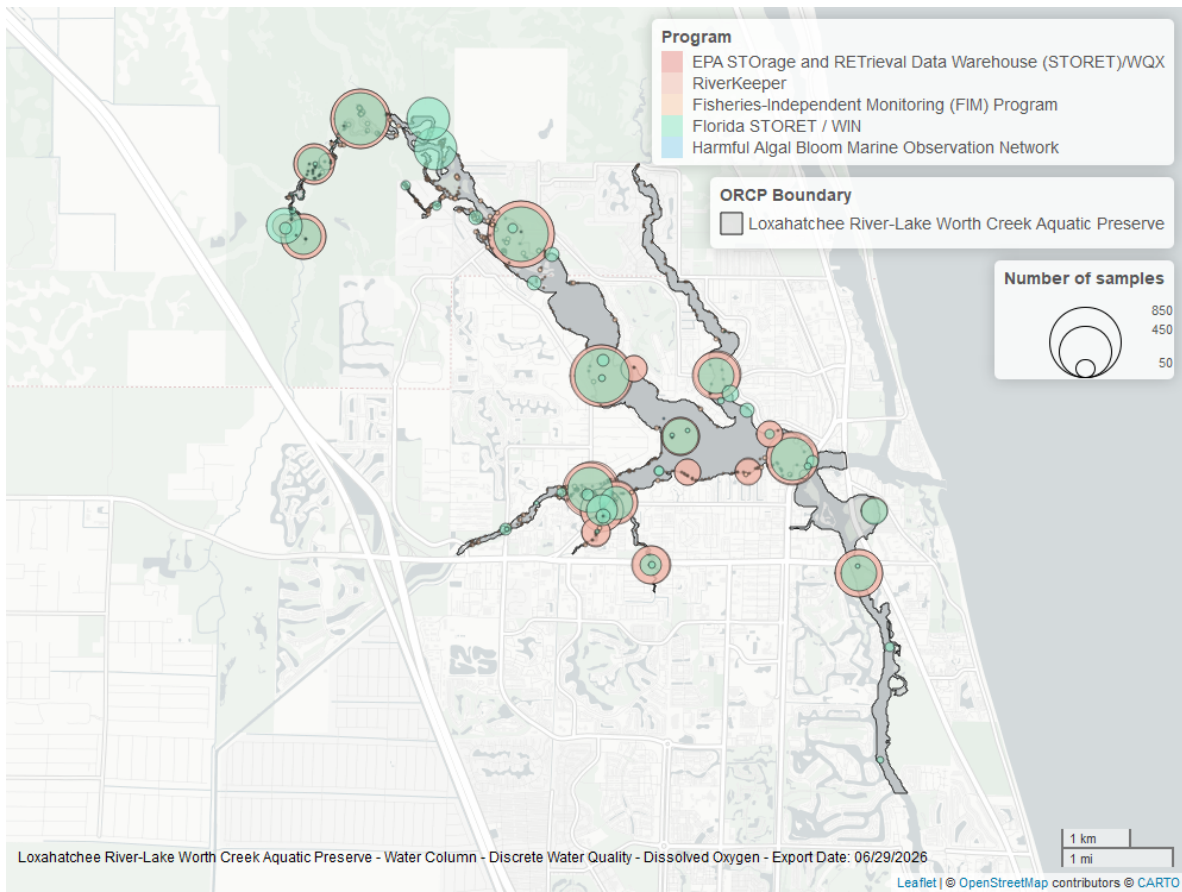


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

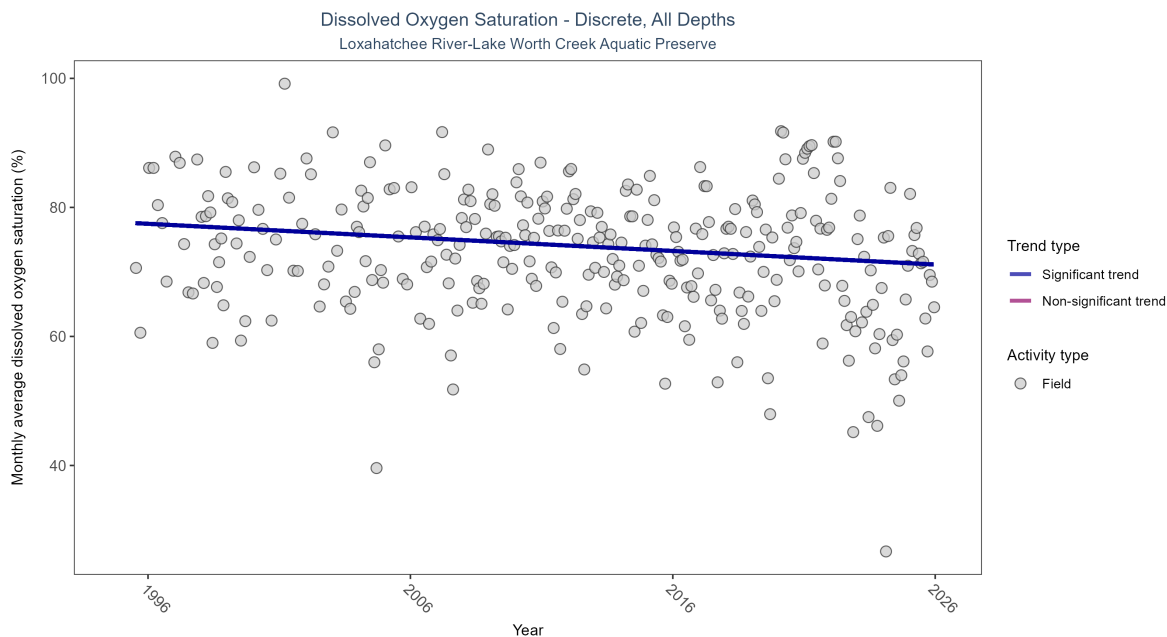


Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation decreased by 0.21% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	11226	31	1995 - 2025	76	-0.15406	77.67006	-0.2105	0.0001

\end{table}

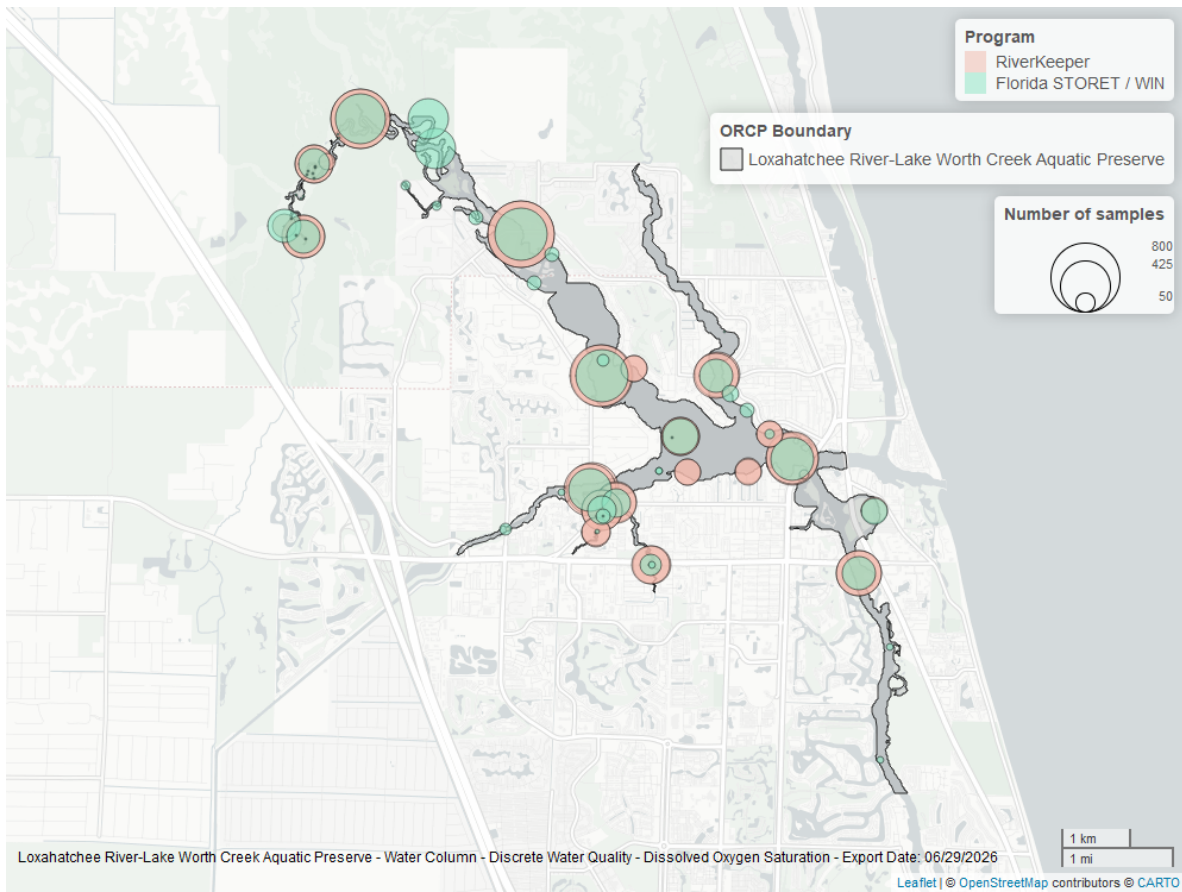


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

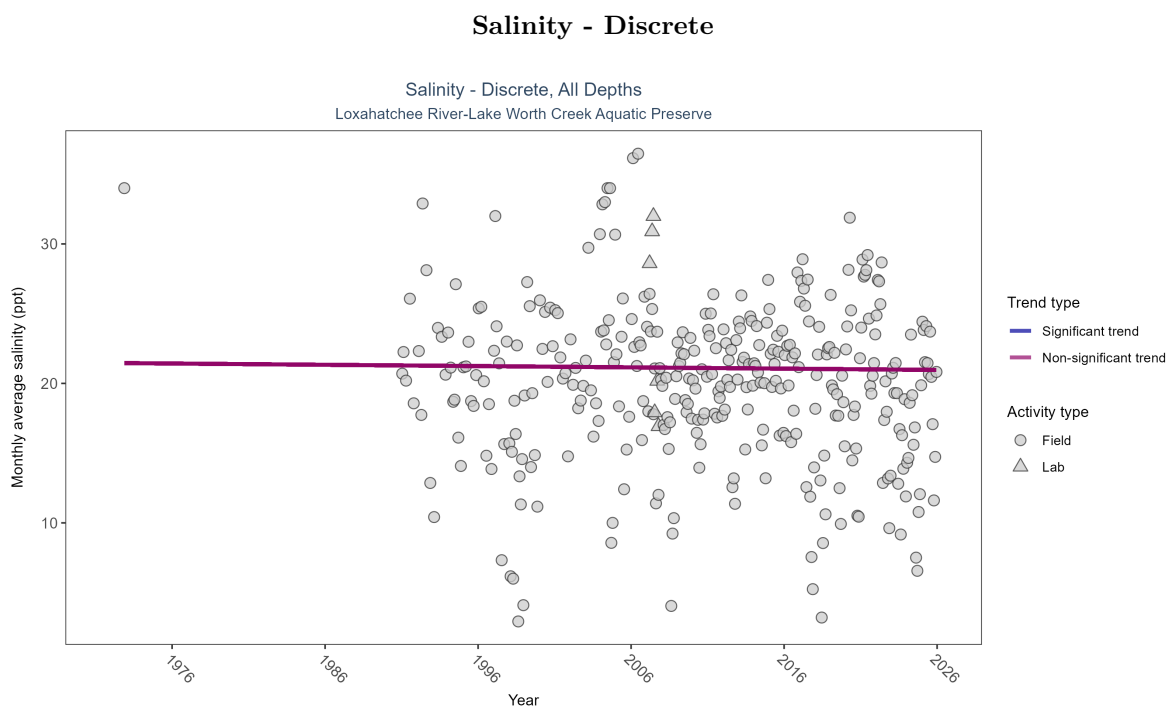


Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 4: Salinity showed no detectable trend between 1972 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	12306	36	1972 - 2025	24.4	-0.00982	21.46318	-0.00921	0.7936

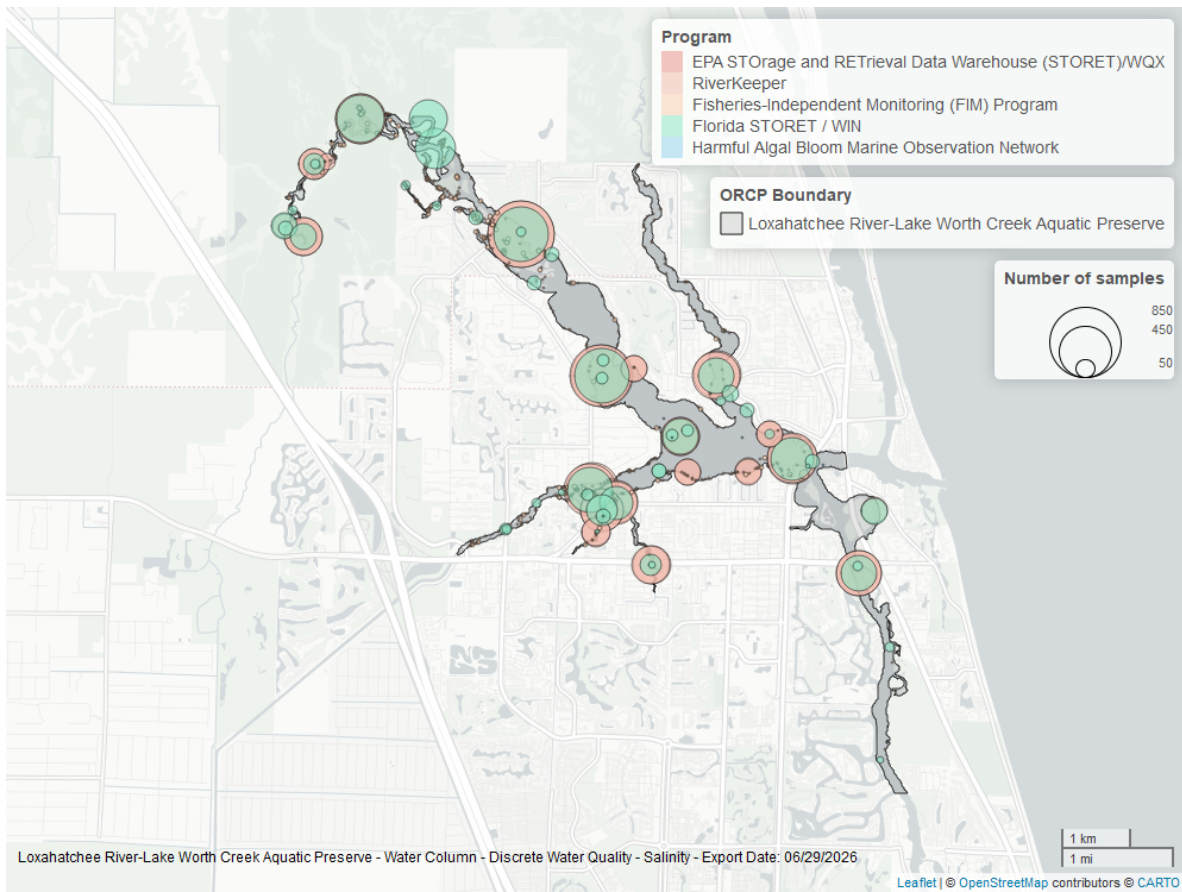


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

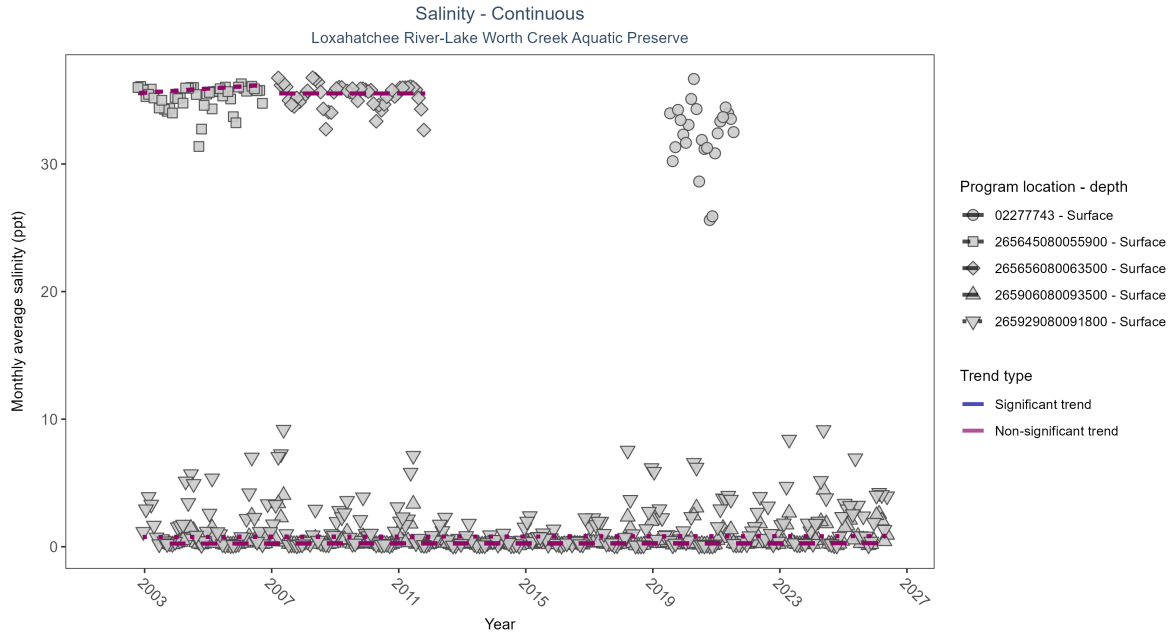


Figure 11: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: No detectable change in monthly average salinity was observed at four locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02277743	Insufficient data	1144	3	2019 - 2021	32.0	-	-	-	-
265645080055900	No detectable trend	1354	5	2002 - 2006	36.0	0.22	35.43	0.16	0.16
265656080063500	No detectable trend	2046	5	2007 - 2011	36.0	0.01	35.53	0	1
265906080093500	No detectable trend	9690	24	2003 - 2026	0.3	0.05	0.25	0	0.23
265929080091800	No detectable trend	8362	25	2002 - 2026	0.4	0.04	0.76	0	0.33

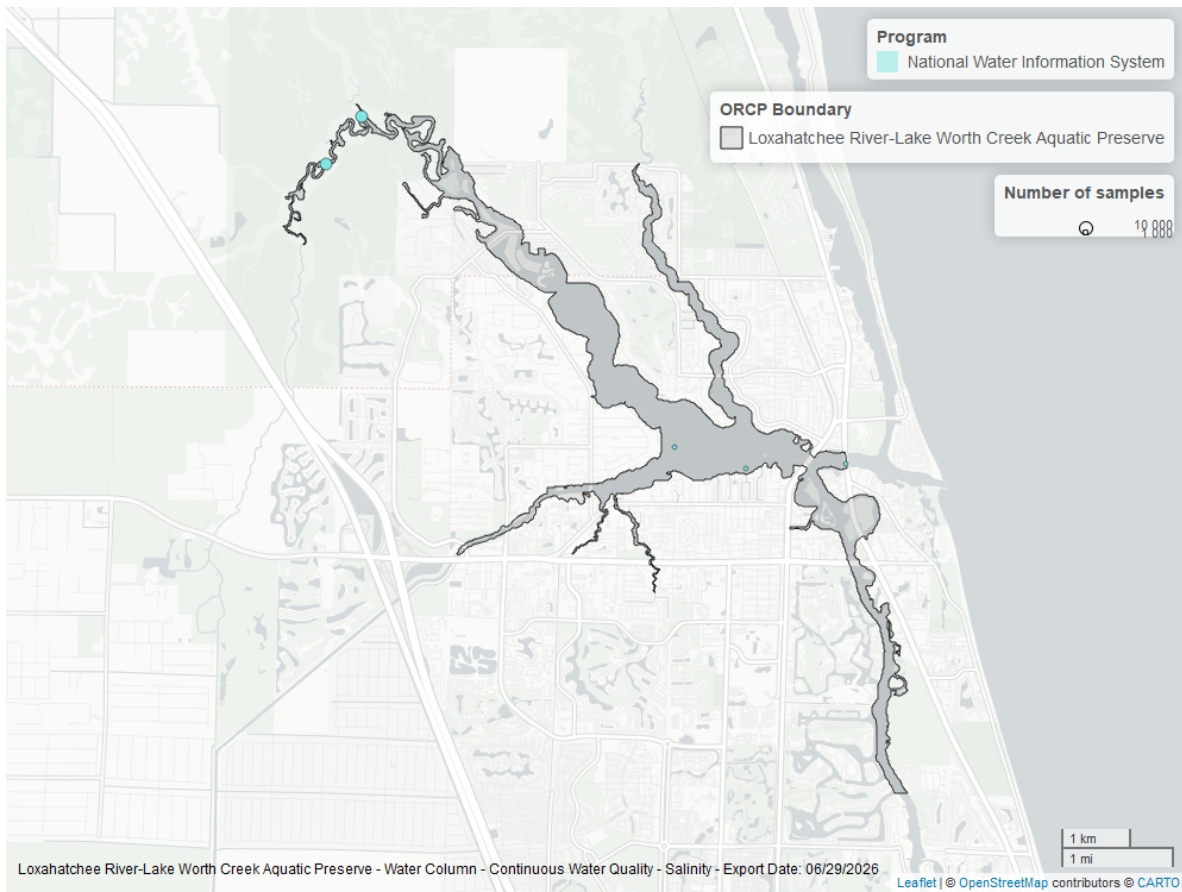


Figure 12: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

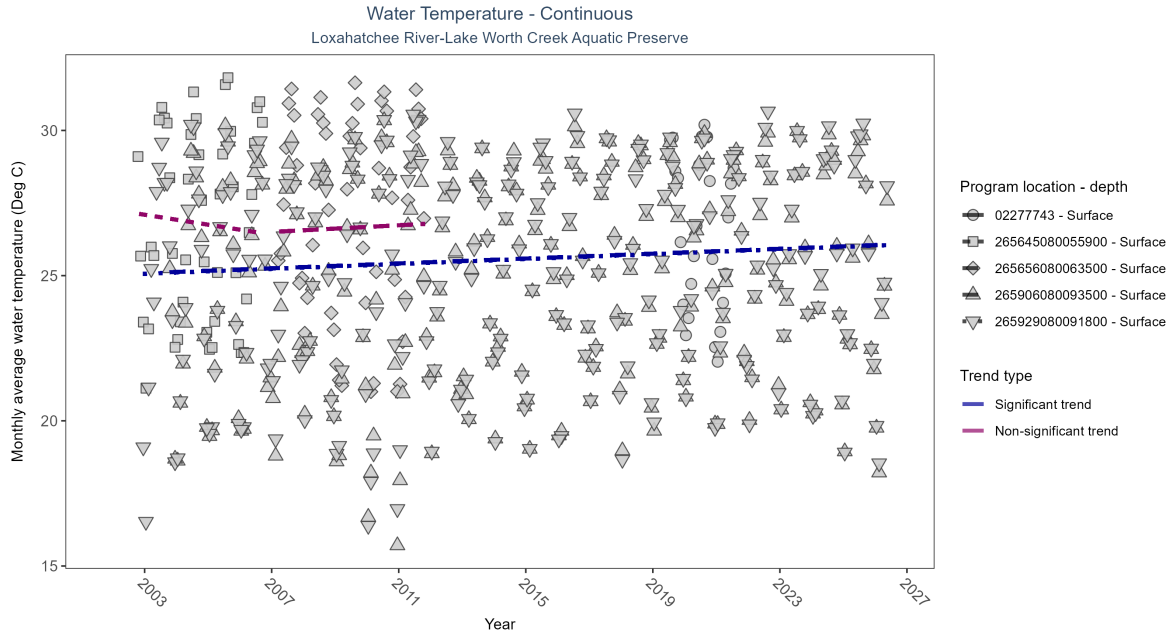


Figure 13: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: At two program locations, monthly average water temperature increased by 0.04Deg C per year. No detectable change in monthly average water temperature was observed at two locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02277743	Insufficient data	1122	3	2019 - 2021	26.3	-	-	-	-
265645080055900	No detectable trend	1354	5	2002 - 2006	26.6	-0.29	27.27	-0.17	0.07
265656080063500	No detectable trend	2664	5	2007 - 2011	27.6	0.08	26.5	0.06	0.45
265906080093500	Increasing trend	12760	24	2003 - 2026	25.6	0.26	25.08	0.04	0
265929080091800	Increasing trend	8383	25	2002 - 2026	25.7	0.23	25.02	0.04	0

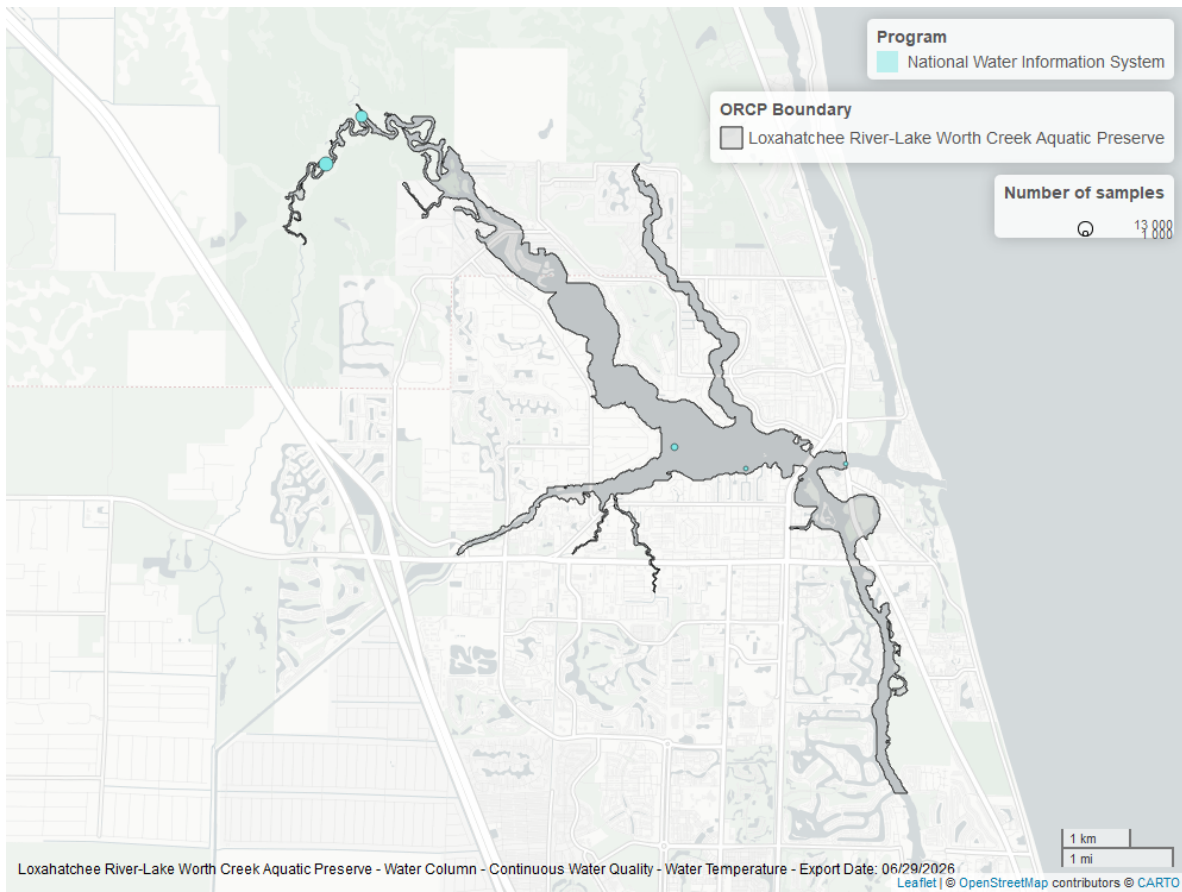


Figure 14: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

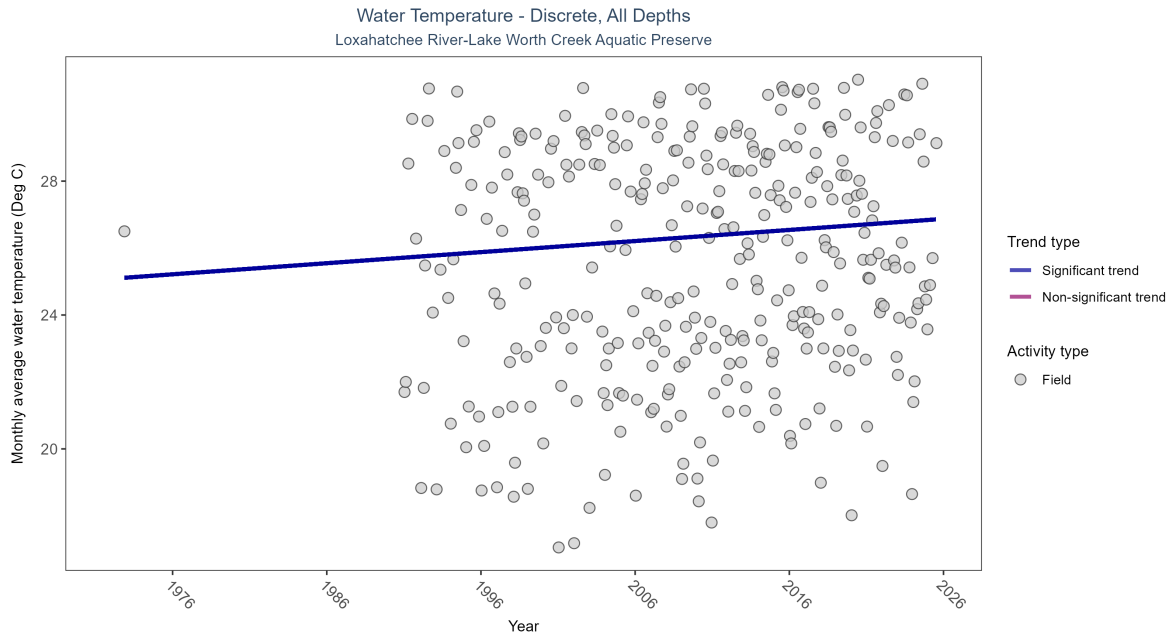


Figure 15: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 7: Monthly average water temperature increased by 0.03Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	10905	36	1972 - 2025	25.6	0.15429	25.08312	0.03314	0.0001

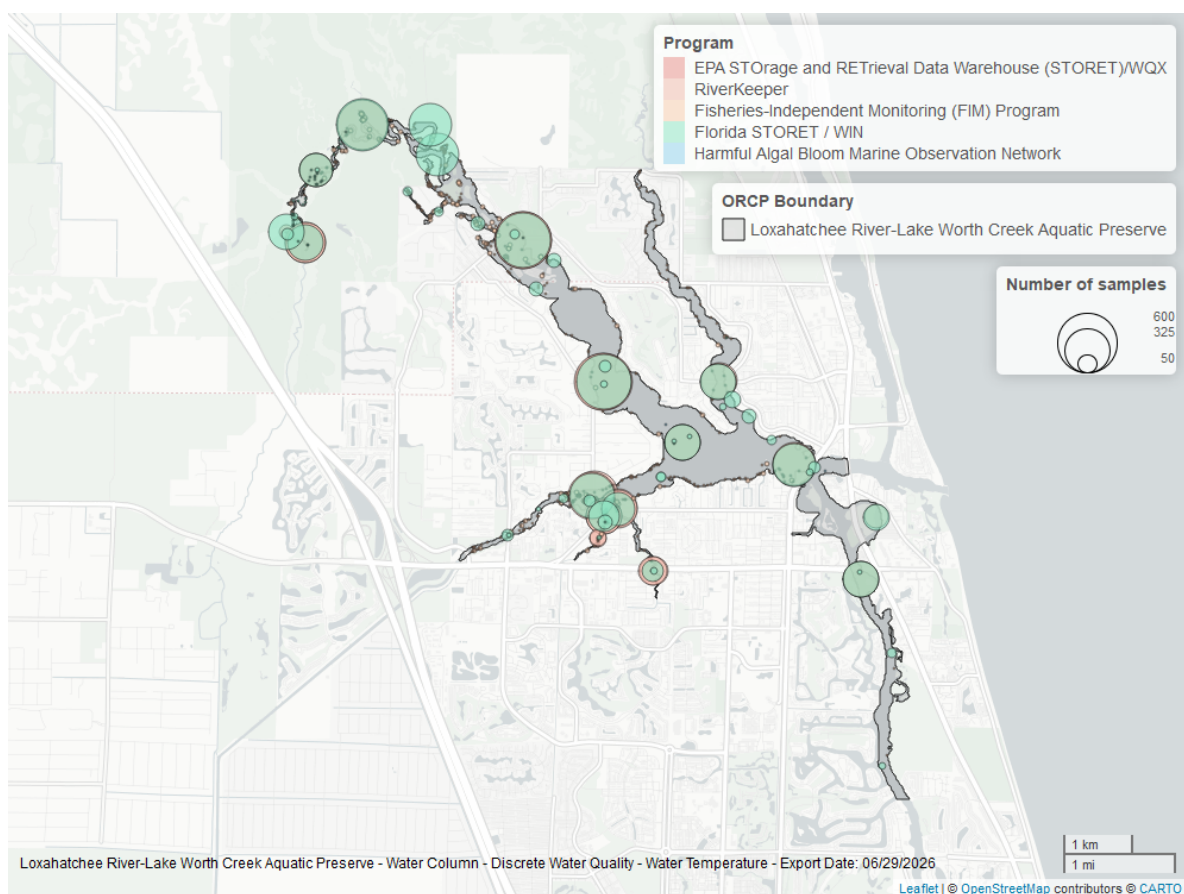


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

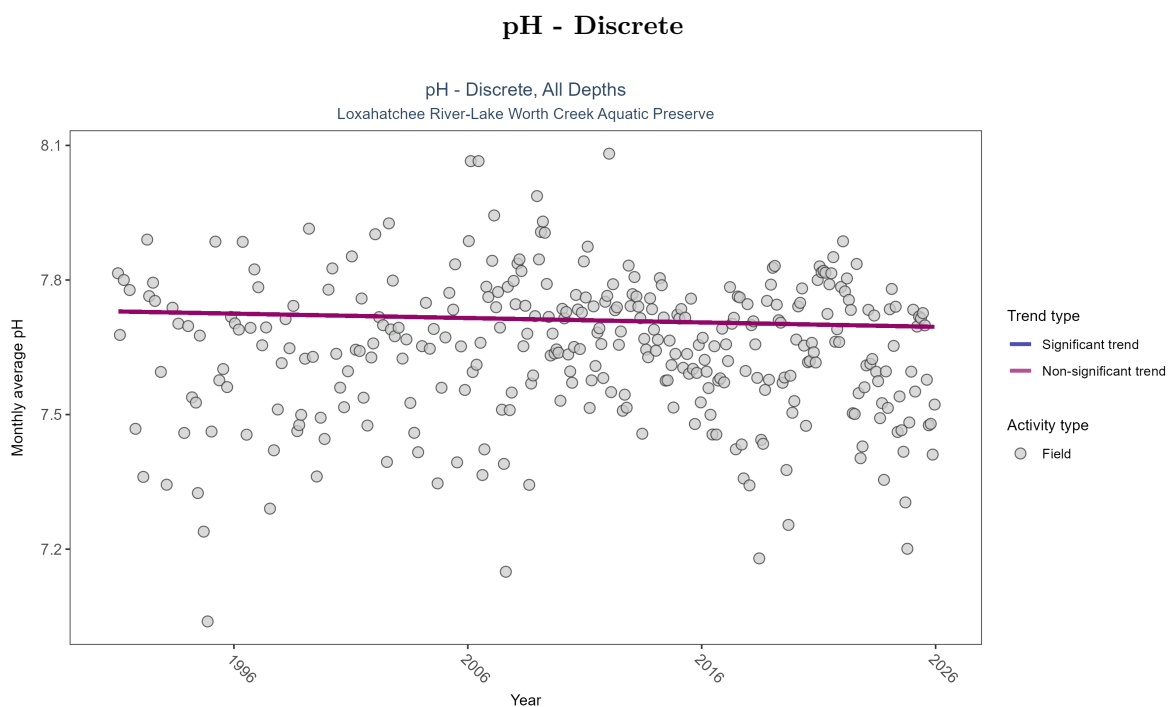


Figure 17: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 8: pH showed no detectable trend between 1991 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	12916	35	1991 - 2025	7.7	-0.06325	7.72989	-0.00098	0.2676

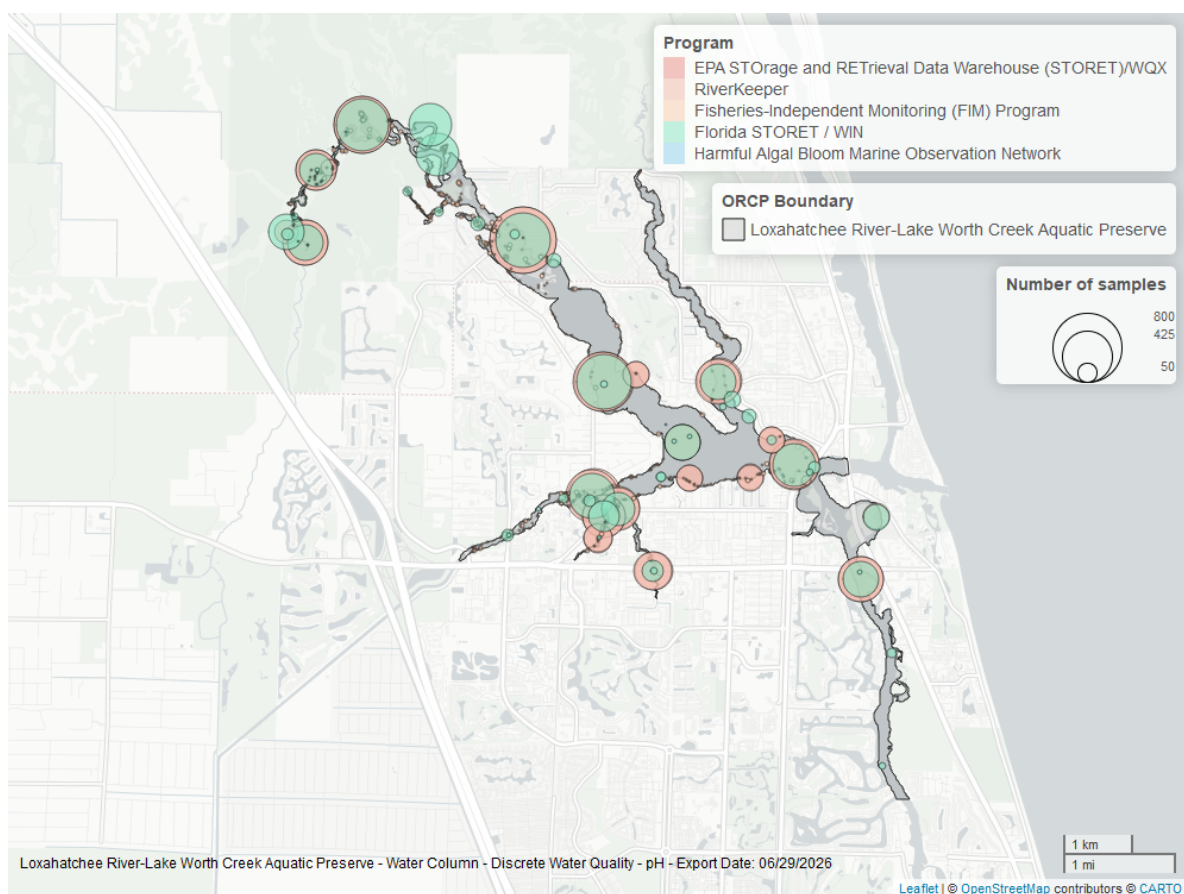


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

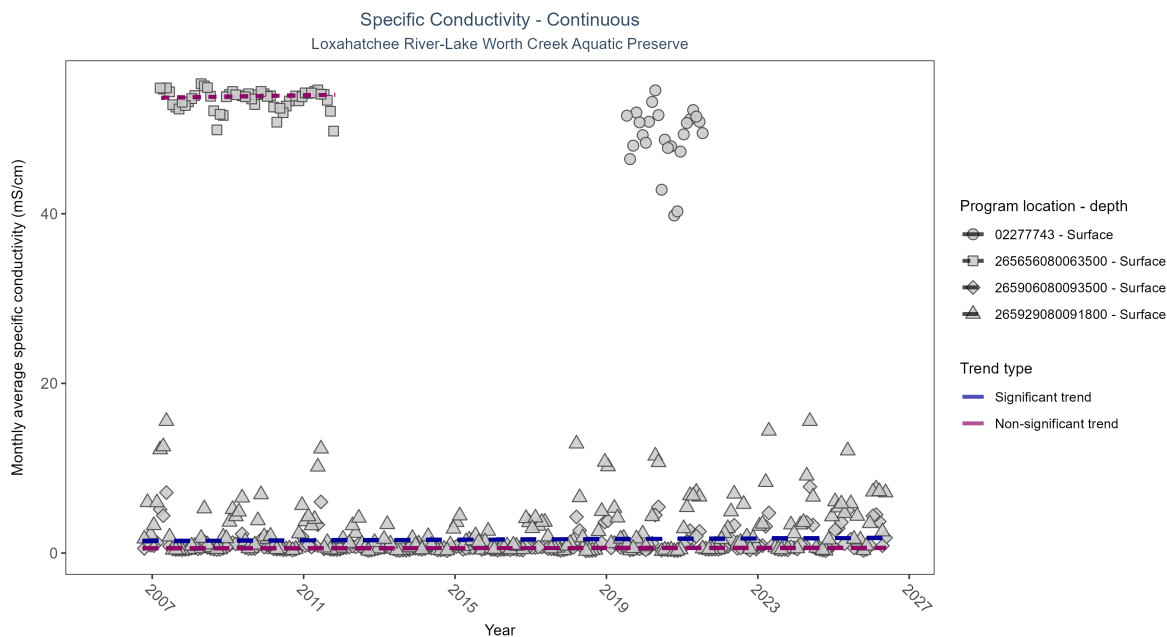


Figure 19: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At one program location, monthly average specific conductivity increased by 0.02 mS/cm per year. No detectable change in monthly average specific conductivity was observed at two locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02277743	Insufficient data	1425	3	2019 - 2021	50.10	-	-	-	-
265656080063500	No detectable trend	2550	5	2007 - 2011	53.90	0.04	53.66	0.08	0.93
265906080093500	No detectable trend	13751	21	2006 - 2026	0.56	0.07	0.55	0	0.14
265929080091800	Increasing trend	6917	21	2006 - 2026	0.85	0.11	1.41	0.02	0.02

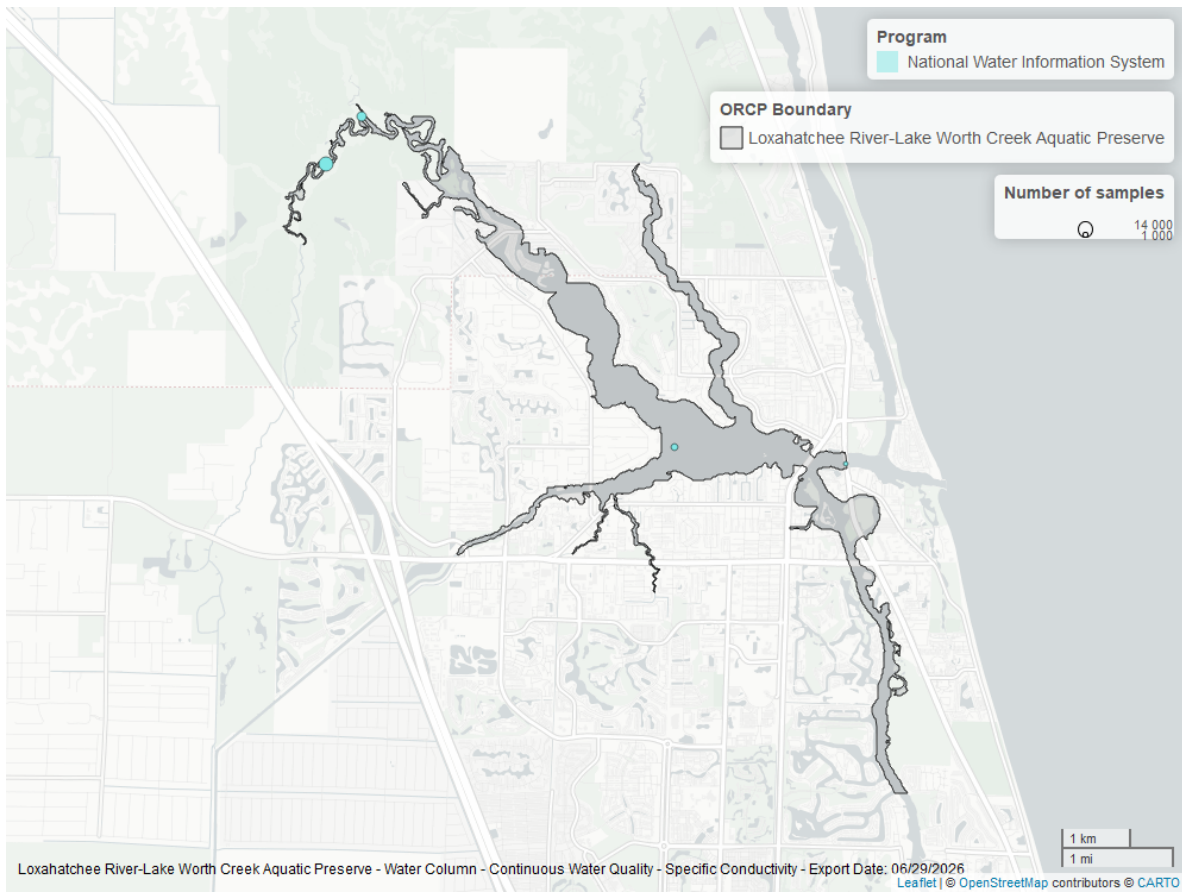


Figure 20: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Discrete

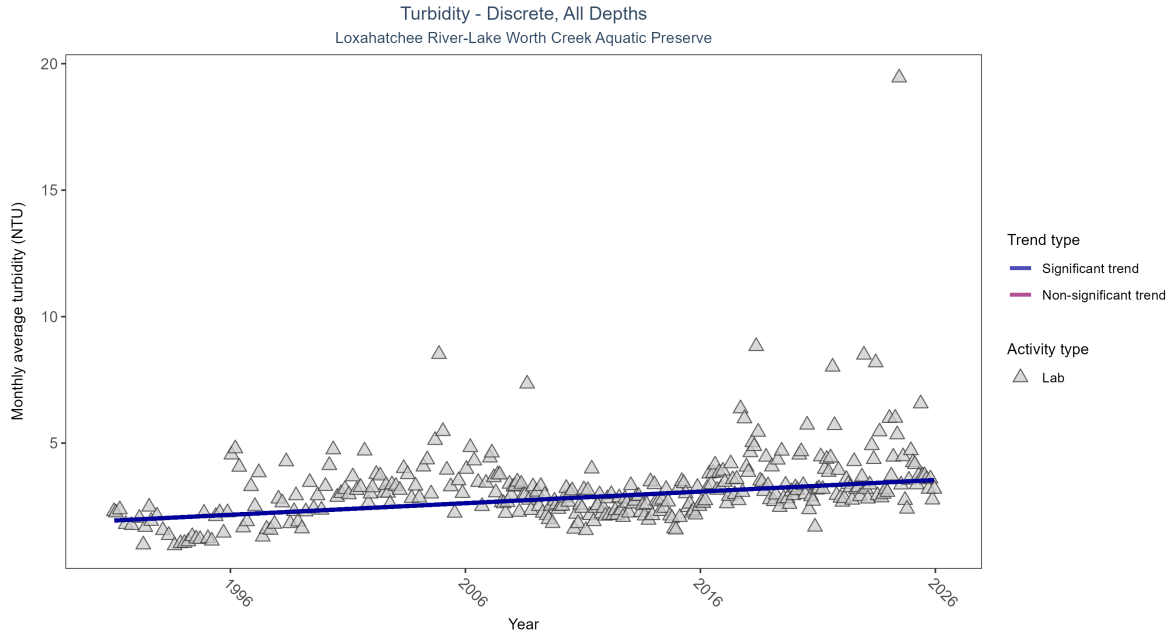


Figure 21: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 10: Monthly average turbidity increased by 0.05 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6632	35	1991 - 2025	2.7	0.3358	1.93661	0.04586	0

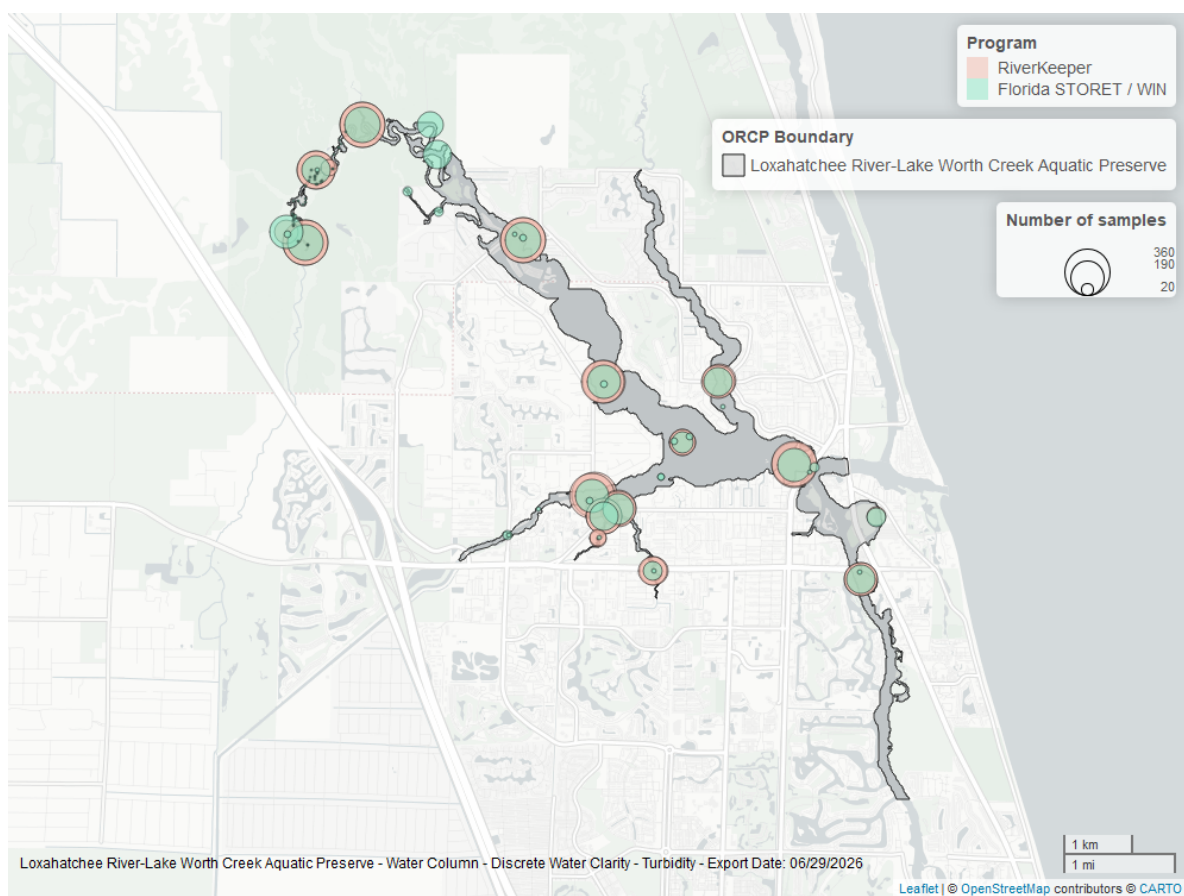


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

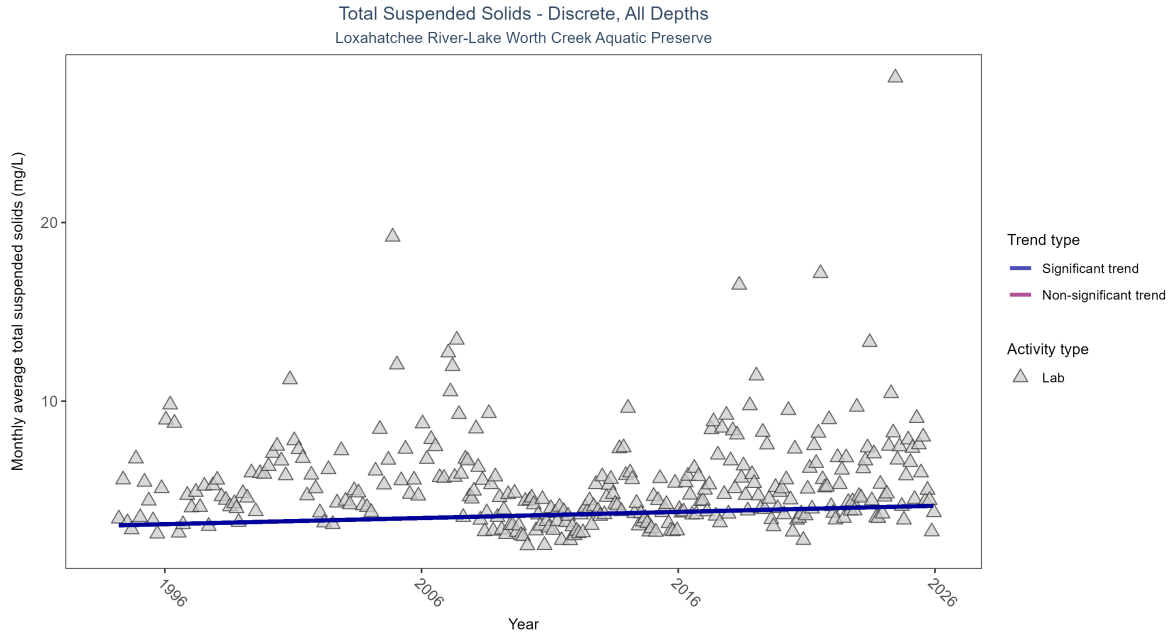


Figure 23: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 11: Monthly average total suspended solids increased by 0.03 mg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6264	32	1994 - 2025	4	0.14544	3.0446	0.03434	0.0056

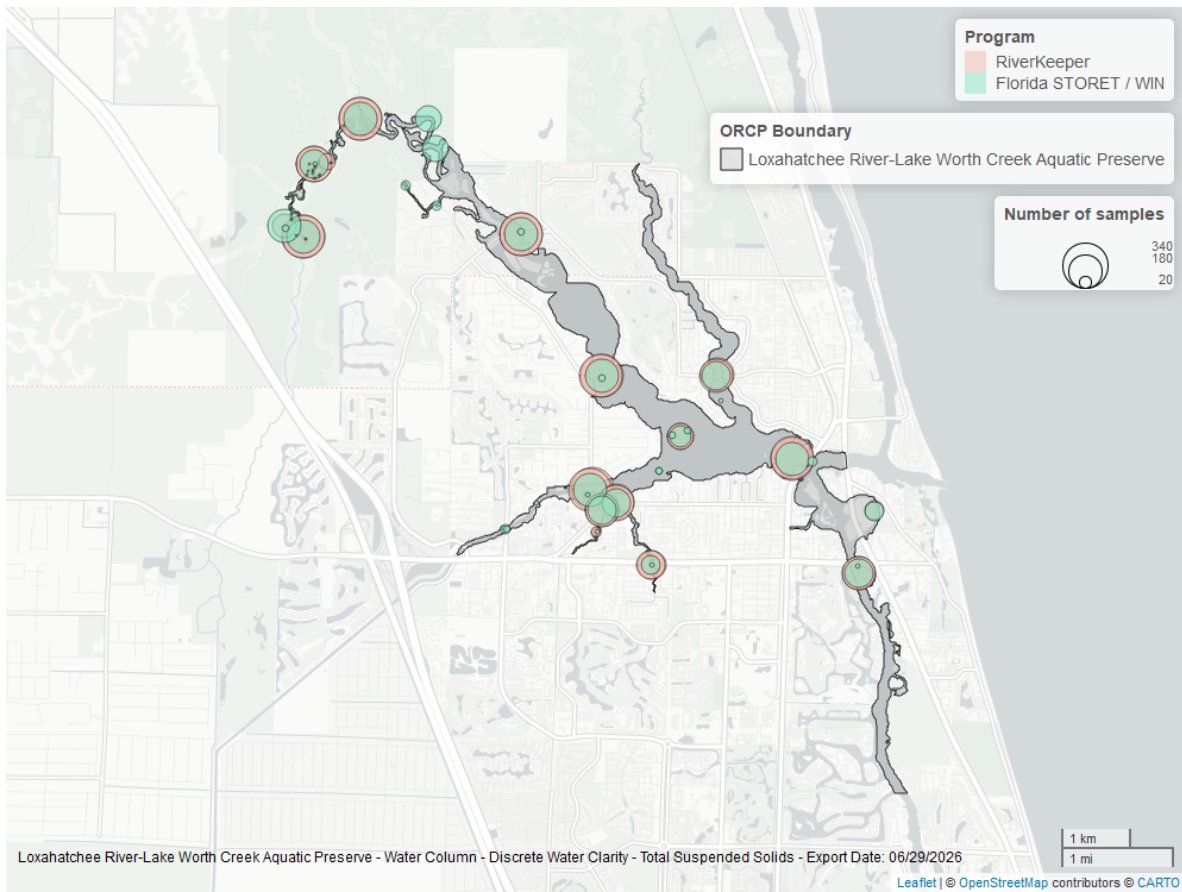


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

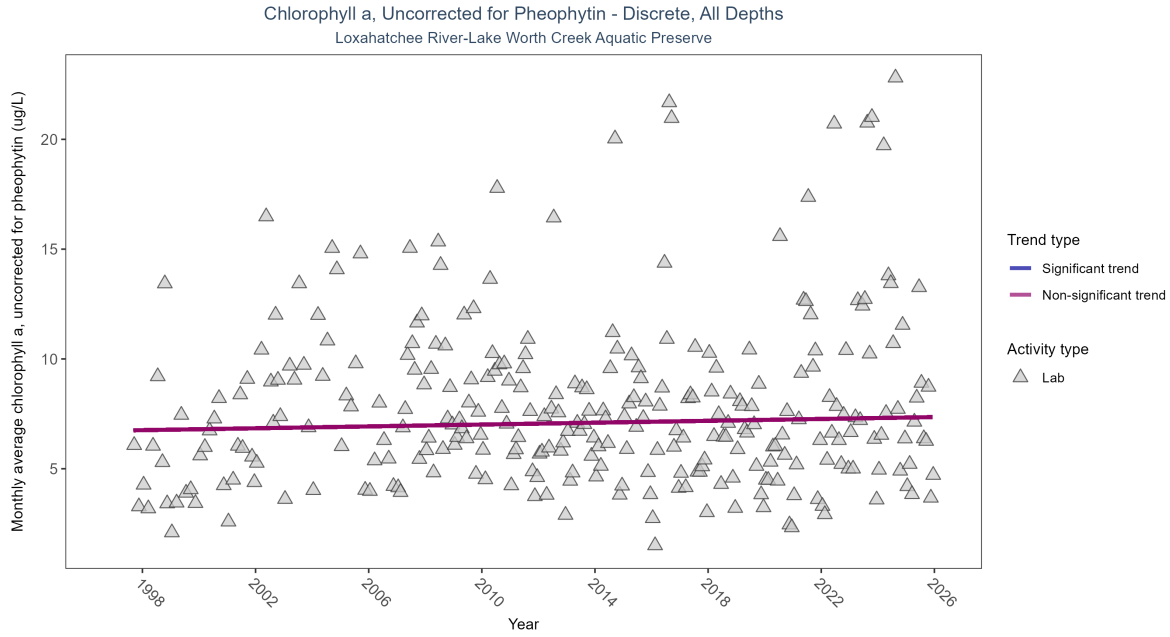


Figure 25: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 12: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1997 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6007	29	1997 - 2025	5.6	0.03611	6.73567	0.02125	0.3828

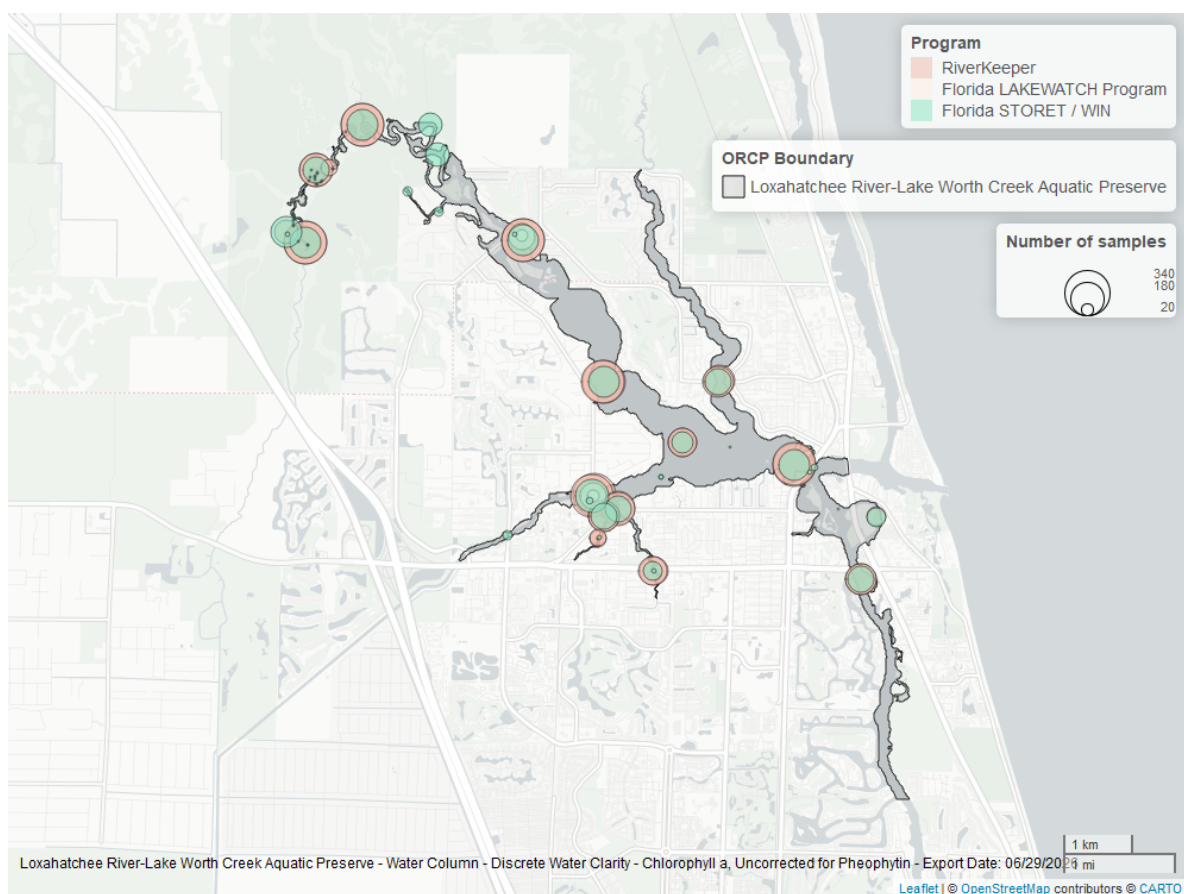


Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

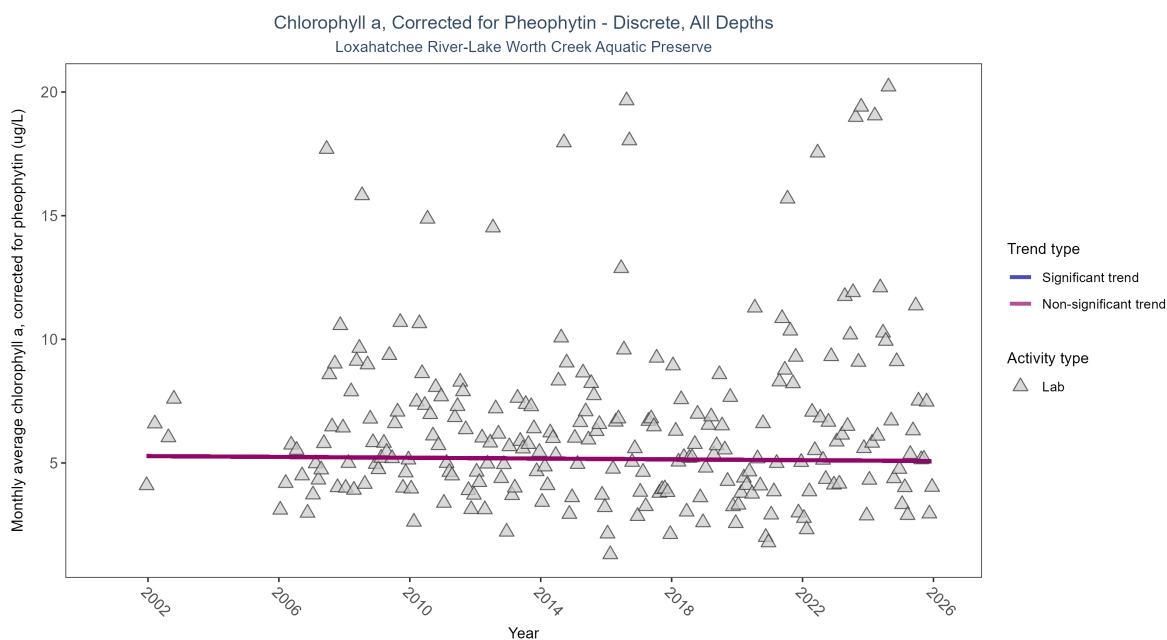


Figure 27: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 13: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2001 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	4638	22	2001 - 2025	4.5	-0.01408	5.28487	-0.00825	0.7474

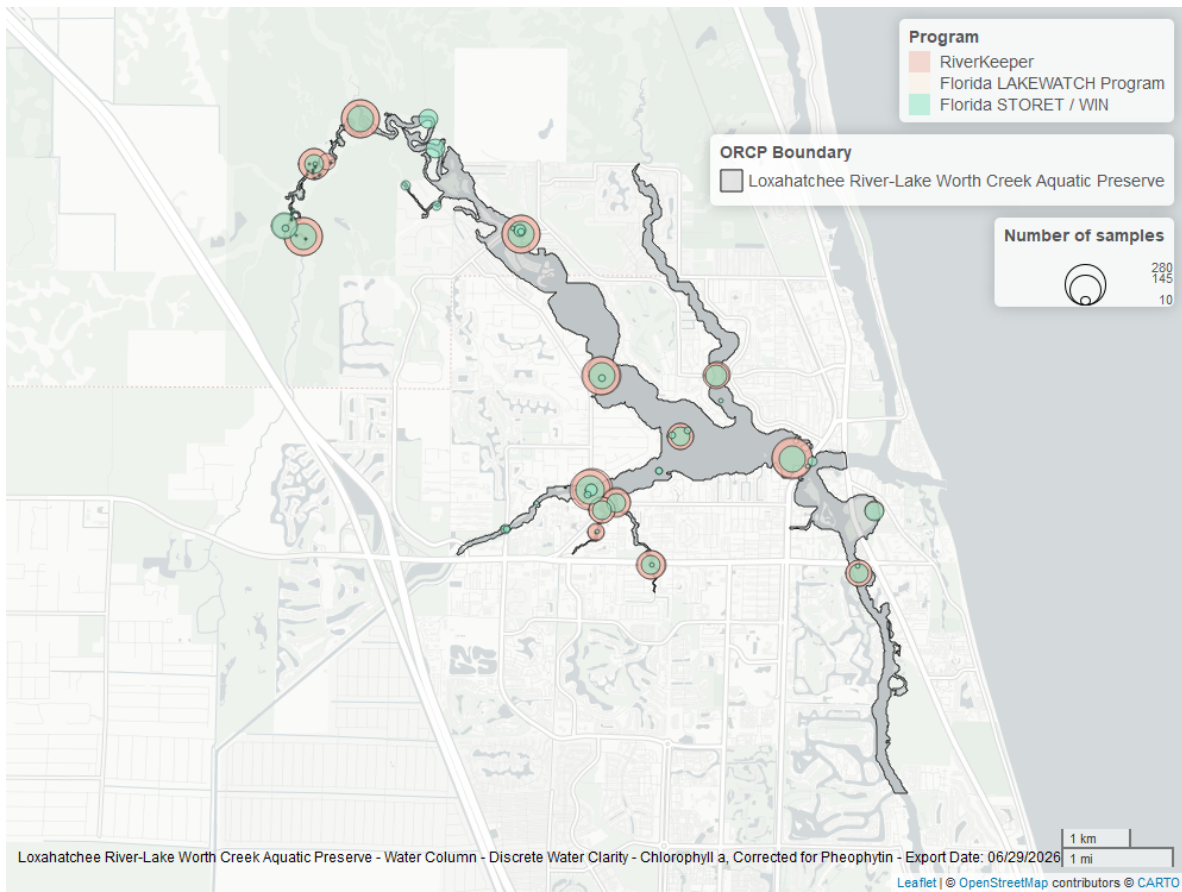


Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

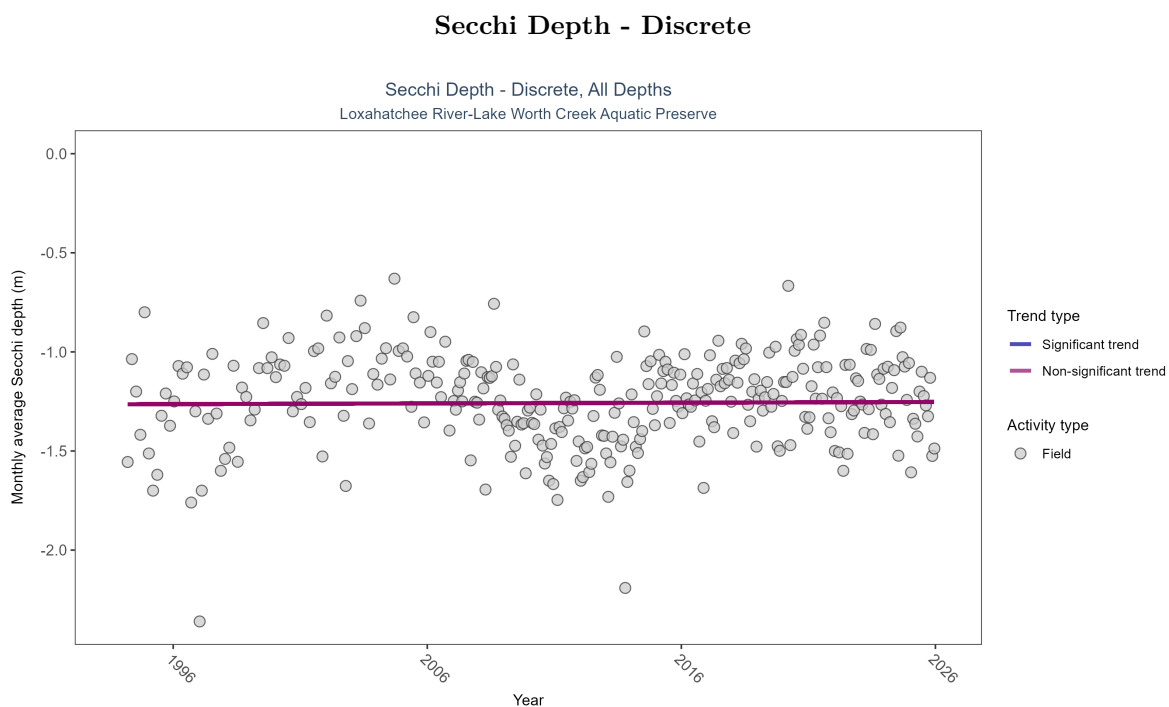


Figure 29: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 14: Secchi depth showed no detectable trend between 1994 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	5062	32	1994 - 2025	-1.2	0.04058	-1.26426	0.00036	0.8485

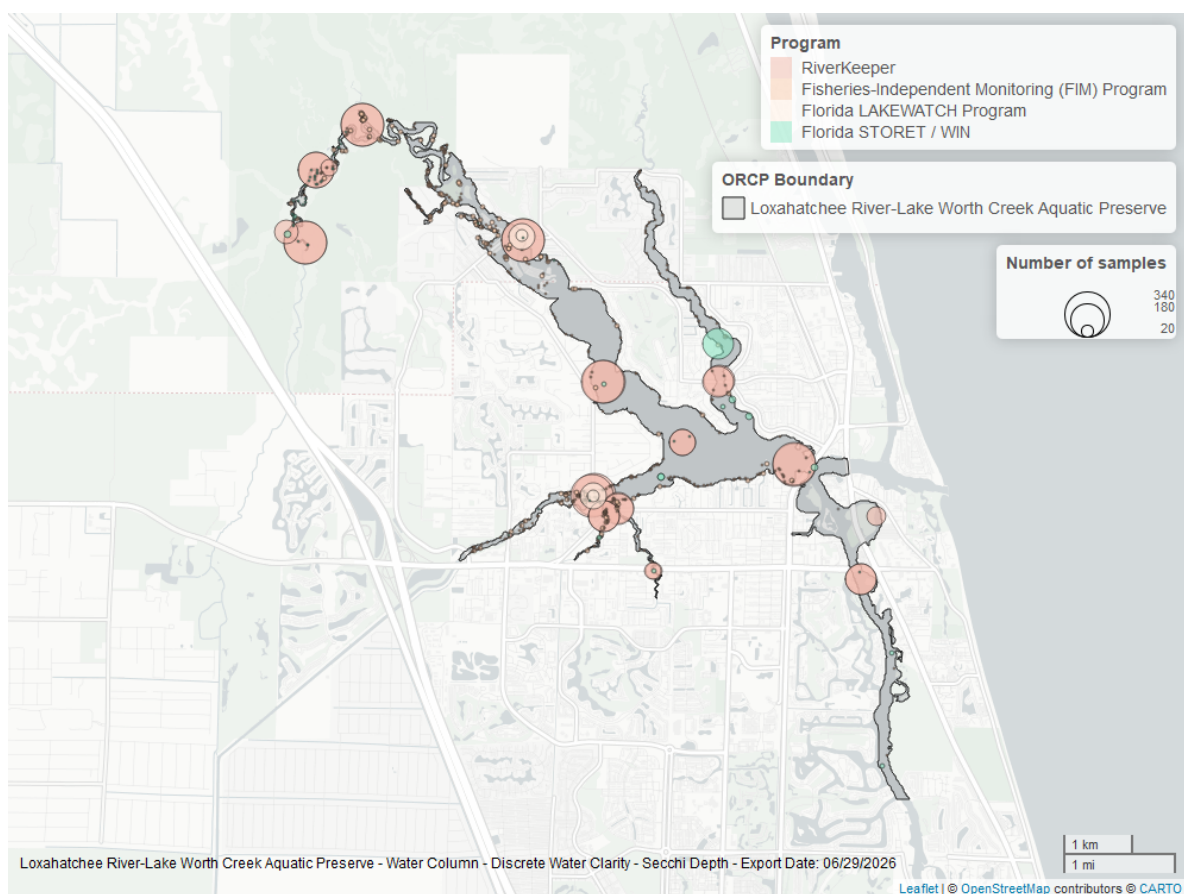


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

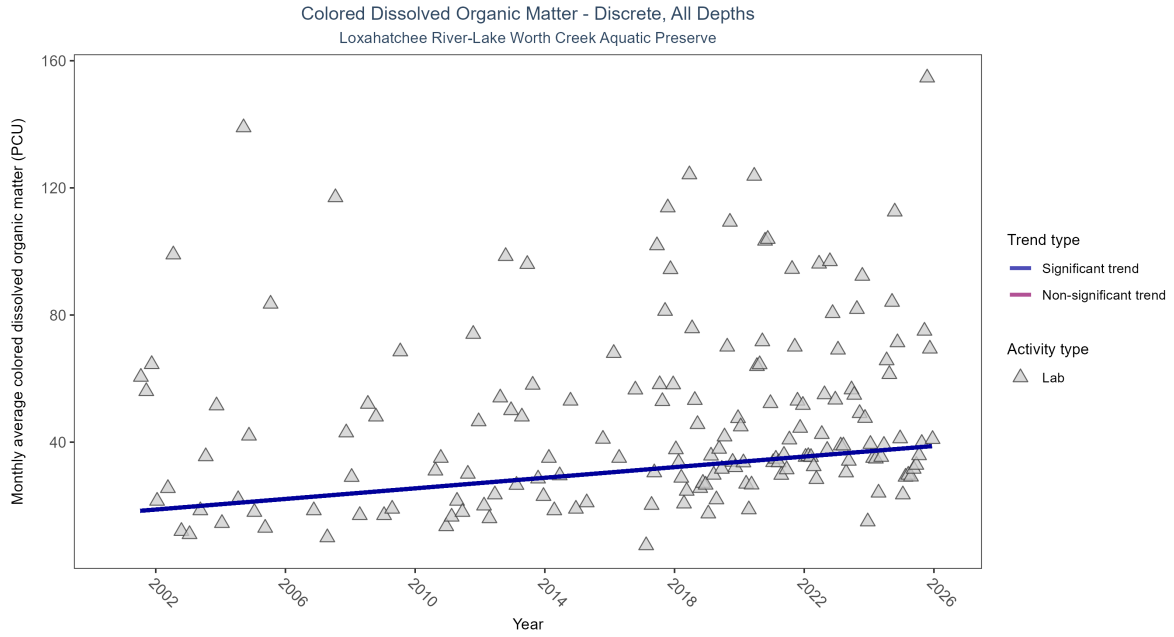


Figure 31: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 15: Monthly average colored dissolved organic matter increased by 0.83 PCU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1291	25	2001 - 2025	45	0.25694	17.96211	0.83333	0

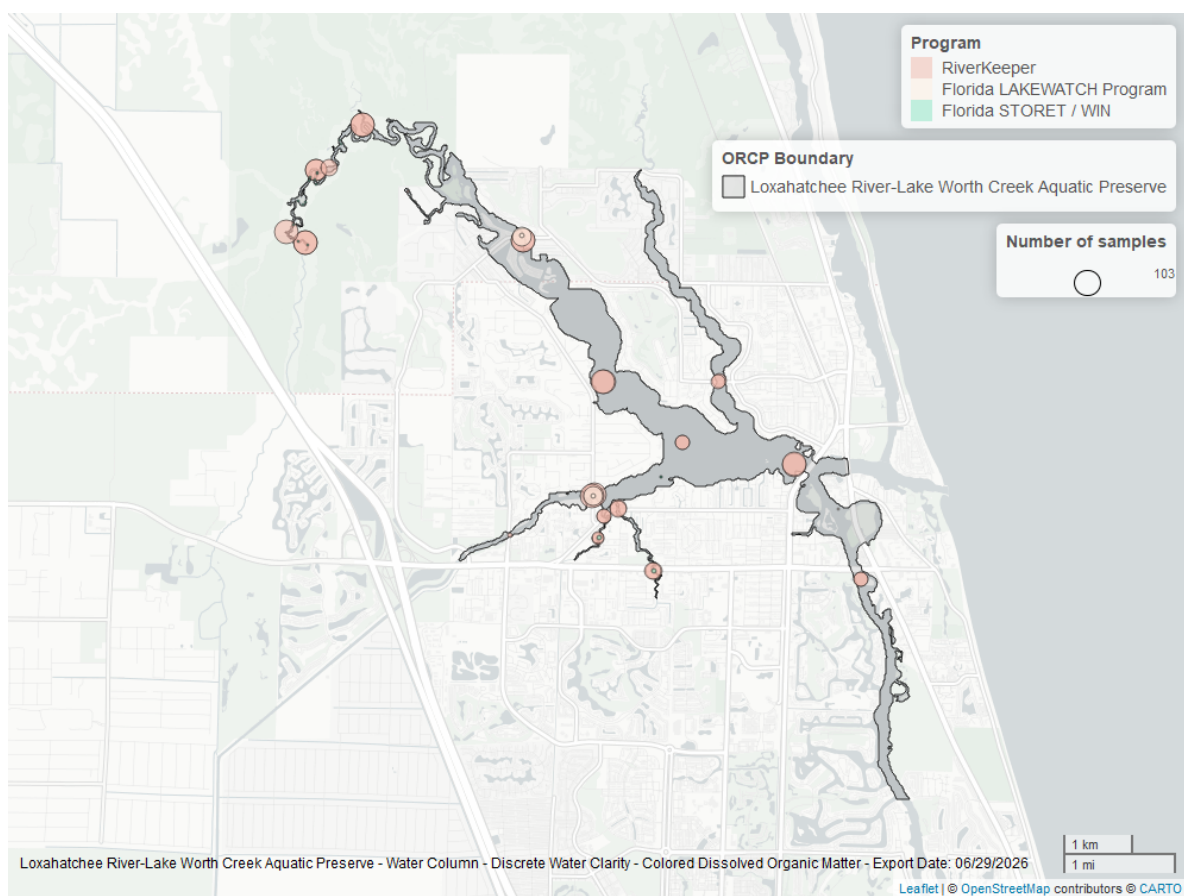


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.